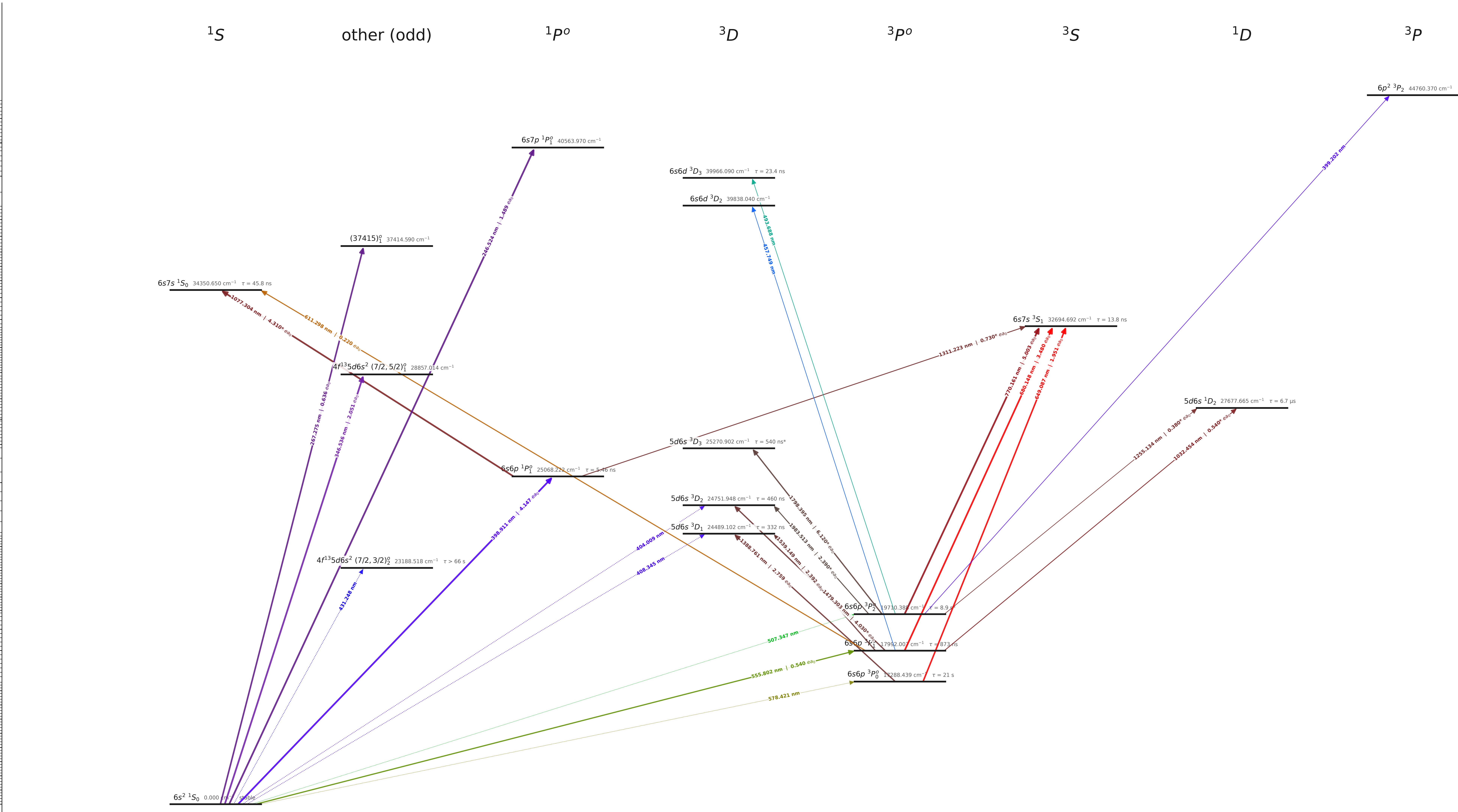


¹⁷³Yb (Yb I), nuclear spin $I = 5/2$. Energy levels and transitions below 2 μm : 18 levels, 26 lines.
Yb⁺ ionisation limit 50443.20 cm^{-1} = 6.25416 eV (198.243 nm)

¹⁷³Yb (Yb I), nuclear spin $I = 5/2$. Energy levels and transitions below 2 μm : 18 levels, 26 lines.

Yb⁺ ionisation limit 50443.20 cm⁻¹ = 6.25416 eV (198.243 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle f || er || i \rangle$ in ea_0
 Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists)



transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \mathbf{r} J\rangle $ ^a	τ (ea)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$6s^2\ ^1S_0 - 6s7p\ ^1P_1^o$	246.5242	246.4497	1216.0772	1.489	1.000e+08	-	-	-	246.5 nm
$6s^2\ ^1S_0 - (37415)_2^o$	267.2754	267.1960	1121.6612	0.636	1.430e+07	-	-	-	267.3 nm
$6s^2\ ^1S_0 - 4f^{13}5d6s^2\ (7/2, 5/2)_2^o$	346.5362	346.4370	865.1115	2.051	6.830e+07	-	-	-	346.5 nm
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.9114	398.7986	751.5264	4.147	1.830e+08	29.13	100	399 nm: Zeeman slower, blue MOT, imaging	
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.0086	403.8945	742.0447	-	-	0.346	-	-	404 nm forbidden transition
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.3449	408.2296	734.1648	-	-	0.4794	-	-	408 nm highly forbidden
$6s^2\ ^1S_0 - 4f^{13}5d6s^2\ (7/2, 3/2)_2^o$	431.2479	431.1266	695.1743	-	-	2.411e-09	-	-	431 nm inner-shell clock transition
$6s^2\ ^1S_0 - 6s6p\ ^3P_2^o$	507.3467	507.2053	590.9026	-	1.120e-01	1.788e-08	99.7	507 nm M2 transition; shelving into 3P2	
$6s^2\ ^1S_0 - 6s6p\ ^3P_1^o$	555.8024	555.6480	539.3868	0.540	1.146e+06	0.1824	100	556 nm: green MOT, narrow-line cooling, imaging in tweeze...	
$6s^2\ ^1S_0 - 6s6p\ ^3P_0^o$	578.4212	578.2609	518.2944	-	4.800e-02	7.579e-09	100	578 nm optical clock transition	
$6s6p\ ^3P_1^o - 6s7s\ ^1S_0$	611.2976	611.1285	490.4198	0.220	4.293e+05	3.475	1.97	611 nm	
$6s6p\ ^3P_0^o - 6s7s\ ^3S_1$	649.0871	648.9078	461.8678	1.951	9.400e+06	11.53	13	649 nm clock-state repumper / 3P0 detection	
$6s6p\ ^3P_1^o - 6s7s\ ^3S_1$	680.1479	679.9602	440.7754	3.480	2.600e+07	11.53	36	680 nm decay channel of 3S1 during repumping	
$6s6p\ ^3P_2^o - 6s7s\ ^3S_1$	770.1606	769.9487	389.2596	5.003	3.700e+07	11.53	50.8	770 nm 3P2 repumper	
$6s6p\ ^3P_1^o - 5d6s\ ^1D_2$	1032.4544	1032.1715	290.3687	0.540 ^a	1.074e+05	0.02375	71.9	1032 nm	
$6s6p\ ^3P_1^o - 6s7s\ ^1S_0$	1077.3043	1077.0093	278.2802	4.310 ^a	3.010e+07	3.475	100	1077 nm main decay channel of 6s7s 1S0	
$6s6p\ ^3P_0^o - 5d6s\ ^1D_2$	1255.1340	1254.7907	238.8530	0.380 ^a	2.959e+04	0.02375	19.8	1255 nm	
$6s6p\ ^3P_1^o - 6s7s\ ^3S_1$	1331.1226	1310.8641	228.6358	0.730 ^a	1.596e+05	11.53	0.21	1311 nm weak decay channel of 3S1	
$6s6p\ ^3P_0^o - 5d6s\ ^3D_1$	1388.7610	1388.3814	215.8705	2.759	1.920e+06	0.4794	63.8	1389 nm clock-state repumper	
$6s6p\ ^3P_1^o - 5d6s\ ^3D_2$	1479.3029	1478.8986	202.6579	4.030 ^a	2.033e+06	0.346	93.5	1480 nm	
$6s6p\ ^3P_1^o - 5d6s\ ^3D_1$	1539.1494	1538.7289	194.7780	2.392	1.060e+06	0.4794	35.2	1539 nm decay channel of 3D1	
$6s6p\ ^3P_2^o - 5d6s\ ^3D_2$	1798.3949	1797.9039	166.7000	6.120 ^a	1.864e+06	0.2947	100	1798 nm cycling transition from 3P2	
$6s6p\ ^3P_2^o - 5d6s\ ^3D_3$	1983.5130	1982.9717	151.1422	2.390 ^a	2.966e+05	0.346	13.6	1983 nm	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	168-174	1887.64(23)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	170-174	1187.8(72)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	171-174	940.8(9)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	172-174	533.0(3)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	173-174	291.51(5)
$6s^2\ ^1S_0 - 6s6p\ ^1P_1^o$	398.911	176-174	-509.25(7)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	170-174	2531(5)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	171-174	2019(6)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	172-174	1100(4)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	173-174	609(6)
$6s^2\ ^1S_0 - 5d6s\ ^3D_2$	404.009	176-174	-1059(11)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	170-174	2542(12)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	171-174	2038.9(12)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	172-174	1112(8)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	173-174	619.4(8)
$6s^2\ ^1S_0 - 5d6s\ ^3D_1$	408.345	176-174	-1063(6)
$6s^2\ ^1S_0 - 4f3s5d6s^2\ (7/2, 3/2)_2^o$	431.248	171-176	-3406.79(3)
$6s^2\ ^1S_0 - 4f3s5d6s^2\ (7/2, 3/2)_2^o$	431.248	173-176	-1797.80(26)
$6s^2\ ^1S_0 - 4f3s5d6s^2\ (7/2, 3/2)_2^o$	431.248	174-176	-1115.787(25)
$6s^2\ ^1S_0 - 4f3s5d6s^2\ (7/2, 3/2)_2^o$	431.248	172-176	-2296.403(25)
$6s^2\ ^1S_0 - 4f3s5d6s^2\ (7/2, 3/2)_2^o$	431.248	170-176	-3926.431(26)
$6s^2\ ^1S_0 - 4f3s5d6s^2\ (7/2, 3/2)_2^o$	431.248	168-176	-5680.383(43)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)					
6s6p $^3P_0^o$	-1094.361(11)	-826.35(8)	F=3/2: +3251.82	F=5/2: +1755.44	F=7/2: -2942.49			
6s6p $^3P_0^o$	-741.98(9)	1347.3(9)	F=1/2: +6136.97	F=3/2: +4417.72	F=5/2: +1889.12	F=7/2: -943.59	F=9/2: -3373.07	
4f ¹³ sd6s ² (7/2, 3/2) ₂ ^o	-309.449(20)	-1700.63(22)	F=1/2: +975.70	F=3/2: +1276.81	F=5/2: +1353.51	F=7/2: +568.04	F=9/2: -1972.40	
5d6s 3D_1	563.50(20)	336.8(3)	F=3/2: -1736.49	F=5/2: -832.94	F=7/2: +1492.95			
6s6p $^1P_1^o$	59.52(20)	601.9(5)	F=3/2: +2112.99	F=5/2: -541.02	F=7/2: +299.27			
6s7s 3S_1	-1891.46(7)	-0.62(16)	F=3/2: +6619.68	F=5/2: +1891.96	F=7/2: -4728.81			
6p ² 3P_0	-123.5(12)	-629.9(1)	F=1/2: +4223.57	F=3/2: +521.77	F=5/2: +527.98	F=7/2: +205.96	F=9/2: -774.98	

Measured A, B constants; sources in the data file